

Initial Project Planning Template

Date	19 September 2026
Team ID	SWTID-2026-2825
Project Name	AI WeatherWise
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

The 15–29 September 2026 build cycle was split into two sprints: **Sprint-1 (19–23 Sept)** covering project setup, database configuration and authentication (Epics 1 & 2), and **Sprint-2 (24–27 Sept)** covering location, weather and AI modules (Epics 3–5), leaving 28–29 Sept for testing, documentation and demonstration.

Sprint	Functional Requirement (Epic)	User Story No.	User Story / Task	Story Points	Priority	Team Member	Sprint Start	Sprint End (Planned)
Sprint-1	Project Setup (Epic 2)	USN-1	As a developer, I can initialize the Node.js project with Express, Mongoose and required dependencies so the backend has a working foundation.	2	High	Safiyah Arif	15 Sep 2026	16 Sep 2026
Sprint-1	Project Setup (Epic 2)	USN-2	As a developer, I can configure environment variables (<code>.env</code> : <code>PORT</code> , <code>MONGO_URI</code> , <code>JWT_SECRET</code> , <code>GEMINI_API_KEY</code>) so secrets stay out of source control.	1	High	Safiyah Arif	16 Sep 2026	16 Sep 2026
Sprint-1	Database Configuration (Epic 4)	USN-3	As a developer, I can connect the app to MongoDB via a reusable <code>db.js</code> module with error handling.	2	High	Safiyah Arif	17 Sep 2026	17 Sep 2026
Sprint-1	Authentication (Epic 3)	USN-4	As a user, I can register for the application by providing my name, email and password (bcrypt-hashed).	2	High	Safiyah Arif	18 Sep 2026	19 Sep 2026
Sprint-1	Authentication (Epic 3)	USN-5	As a user, I can log in with my email & password and receive a JWT access token.	2	High	Safiyah Arif	19 Sep 2026	20 Sep 2026
Sprint-1	Authentication (Epic 3)	USN-6	As a system, I can protect private routes using JWT-verification middleware.	1	High	Safiyah Arif	20 Sep 2026	20 Sep 2026
Sprint-2	Location Management (Epic 3)	USN-7	As a user, I can add a favorite city (with country) to my account.	2	High	Kelwin Benjamin A	21 Sep 2026	21 Sep 2026
Sprint-2	Location Management (Epic 3)	USN-8	As a user, I can view, update and delete my saved favorite locations.	2	High	Kelwin Benjamin A	22 Sep 2026	22 Sep 2026
Sprint-2	Weather Retrieval (Epic 3)	USN-9	As a user, I can fetch real-time weather metrics for any of my saved cities via OpenWeatherMap.	3	High	Clement Anthony Martin	22 Sep 2026	23 Sep 2026
Sprint-2	AI Insights (Epic 3)	USN-10	As a user, I can request an AI-generated natural-language weather summary for a city.	3	High	Firos Shah A M	23 Sep 2026	24 Sep 2026
Sprint-2	AI Insights (Epic 3)	USN-11	As a user, I can request a personalized activity/clothing recommendation based on current weather.	2	Medium	Firos Shah A M	24 Sep 2026	25 Sep 2026
Sprint-2	Reliability (Epic 3)	USN-12	As a system, I can fall back gracefully when the Gemini or OpenWeatherMap key/service is unavailable.	2	Medium	Firos Shah A M	25 Sep 2026	26 Sep 2026
Sprint-2	Testing & Execution (Epic 5)	USN-13	As a QA member, I can verify all endpoints with Thunder Client/Postman and export a <code>postman_collection.json</code> .	2	High	All Members	26 Sep 2026	27 Sep 2026

Total estimated story points: 26. Remaining calendar days (28–29 Sept 2026) were reserved for Performance Testing, Project Documentation, and the final Project Demonstration.